

Caleb O'Neal

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EDUCATION

Bachelor of Science in Computer Engineering University of Virginia	Expected May 2028 GPA: 4.0
Associate of Science in Computer Science Northern Virginia Community College	June 2025 GPA: 3.9

WORK EXPERIENCE

Intern Virginia Tech Qualcomm Thinkabit Lab	August 2024 – May 2025 Falls Church, VA
- Programmed a 3-phase brushless DC motor simulation with a web interface to educate robotics students - Assisted in lab tasks like cleaning, organizing, leading school groups, and moving the lab between buildings	
Teaching Assistant Arlington Career Center	January 2025 – May 2025 Arlington, VA
- Taught a college level statistics class as the primary instructor, covering the principles of statistical analysis - Graded and reviewed students' assignments, including a poster session of students' individual research of a statistical relationship in real world datasets	
High School Intern Department of Defense	June 2024 – August 2024 Virginia
- Redesigned a department website to guide agency management to enterprise architecture visualization tools - Migrated the team's Git repositories between Git hosts, ensuring commit history was preserved - Collaborated with the team's Scrum Master to follow Agile software development best practices	

PROJECTS

PyCon US 2024 Poster Presentation , NASA App Development Challenge	2024
- Developed a 3D interactive simulation of lunar surface exploration for the planned 2027 NASA Artemis III mission to the lunar south pole - Presented our simulation to software industry professionals at the PyCon 2024 poster presentation session - Collaborated with a team to develop the app and manage team finances and logistics	
Martian Crater Data Analysis	
- Analyzed a dataset of 350,000 Martian craters to find a statistical relationship between crater eccentricity and its latitude on Mars, likely as a result of volcanic activity on the Martian surface - Programmed in SAS to perform statistical analysis and generate diagrams showing relevant data - Presented results to peers for review and at a local community college for a larger audience	
Captain, Programming Lead , Robotics Team	August 2022 – May 2025
- Managed a team of 50 with an annual budget of \$50,000 to design, build, wire, program, and test a robot designed for the 2024 and 2025 FIRST Robotics Competitions - Redesigned the team management structure following significant growth in members - Led a team of 10 programmers using GitHub's pull request and issue systems for code review	

SKILLS

- Programming:** Python, C, C++, Java, JavaScript, SAS, Rust, Zig
- Tools:** Git, CI/CD, Linux, KiCad, QEMU